

CSE 309 Theory of Automata, Spring 2026
Exercises 3

1. **DFAs and NFAs** Attempt the Chapter 1 Exercise Questions 1.8 to 1.11 in the class text book Introduction to the Theory of Computation by Michael Sipser, third edition, Pages 84–85.

2. **Constructing DFAs from other DFAs**

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA that accepts some language $L = L(M)$ over the alphabet $\Sigma = \{a, b\}$. For each of the following four languages L' , give the specification of five-tuple $M' = (Q', \Sigma', \delta', q'_0, F')$ such that M' is the specification of a DFA accepting the language L' .

- (a) $L_1 = \overline{L(M)}$, i.e., the complement of L .
- (b) $L_2 = \{a\} \cdot L$
- (c) $L_3 = L \cdot \{a\}$
- (d) $L_4 = L \setminus \{\lambda\}$ - *hint*: Consider setting $Q' = Q \cup \{q_{new}\}$ where $q_{new} \notin Q$ is a new state.

3. **Pumping Lemma for Regular Languages** Here are three incorrect statements of the Pumping Lemma for regular languages. In each case, explain why the statement is *not* correct. Try to identify the most serious problem.

- (a) Suppose L is a regular language over the alphabet Σ . Then the following property must be true. For every string $w \in L$, there exists a constant n such that $|w| \geq n$, and it is possible to write w as $w = xyz$ such that
 - i. $y \neq \lambda$
 - ii. $|xy| \leq n$ and
 - iii. for every integer $k \geq 0$, the string xy^kz is also in L .
- (b) Suppose L is a regular language over the alphabet Σ . Then there exists a constant n and a string $w \in L$ such that $|w| \geq n$, and it is possible to write w as $w = xyz$ such that
 - i. $y \neq \lambda$
 - ii. $|xy| \leq n$ and
 - iii. for every integer $k \geq 0$, the string xy^kz is also in L .
- (c) Assume L is regular. Let n be the constant mentioned in the Pumping Lemma for Regular Languages. Choose a string w such that $w \in L$ and $|w| \geq n$, and get a contradiction by showing that it is not possible to write w as $w = xyz$ such that
 - i. $y \neq \lambda$
 - ii. $|xy| \leq n$ and
 - iii. for every integer $k \geq 0$, the string xy^kz is also in L .

- (d) **Common Mistakes in the Pumping Lemma** The following incorrect proofs show common mistakes in using the pumping lemma. In each case try to find the most serious problems with the argument.

Claim: The language $L = \{a^i b^i \mid i \geq 0\}$ is not a regular language.

- i. **Attempt One.** Suppose that L is regular. Thus the Pumping Lemma for Regular Languages holds. Let n be the constant from the pumping lemma and consider the string $w = a^n b^n$. It is easy to see, by the definition of L , that $w \in L$. Furthermore, $|w| = 2n \geq n$. By the definition of the Pumping Lemma for Regular Languages, there must exist x, y, z such that the last three conditions of the Pumping Lemma are satisfied. Let $x = \lambda, y = a$, and $z = a^{n-1} b^n$. Then $y \neq \lambda$ and $|xy| = |a| = 1 \leq n$, so that the first two conditions for x, y and z are satisfied. However,

$$xy^2z = \lambda a^2 a^{n-1} b^n = a^{n+1} b^n \notin L,$$

so that the third condition is not satisfied. We therefore have a contradiction. It follows that the language L is not regular.

- ii. **Another Attempt.** To reach a contradiction, assume that L is a regular language. Consider $w = a^n b^n$. Clearly $w \in L$ and $|w| \geq n$. Since $|xy| \leq n$, y contains only a 's and $y = a^p$ for some $1 \leq p \leq n$ (since $|y| > 0$). Now consider $xy^2z = a^{n+p} b^n$ and since $p > 0$, $xy^2z \notin L$. We have a contradiction. Hence the assumption that L is regular must be false.